

I • Master Field Equation

Stationary (Helmholtz):

$$(\nabla^2 + k^2) G(x, x') = -\delta^3(x - x')$$

Time-dependent (d'Alembertian):

$$\left(\frac{1}{v_s^2} \partial_t^2 - \nabla^2 + k^2\right) \Phi_{\mu\nu}(x, t) = -J_{\mu\nu}(x, t)$$

Retarded propagator ($\tau = t - t' > 0, r = |x - x'|$):

$$G_R(r, \tau) = \frac{v_s}{4\pi r} e^{-kv_s \tau} \delta(\tau - r/v_s) \theta(\tau)$$

Memory kernel ($\tau_m = 1/kv_s$):

$$K(\tau) = K_0 e^{-\tau/\tau_m} \theta(\tau)$$

General solution (field = integral of past):

$$\Phi(x, t) = \Phi^{(0)} + \int_{-\infty}^t dt' \int d^3x' G_R J$$

II • Type-Theoretic Architecture

Σ -type (soma-field as fiber bundle over Scale):

$$\text{SomaField} \equiv \sum_{(\sigma : \text{Scale}_{20})} \text{Substrate}(\sigma)$$

Zoom Operator (dependent type constructor):

$$\Lambda : (\sigma : \text{Scale}_{20}) \rightarrow \text{Sub}(\sigma) \rightarrow \text{Sub}(\sigma+1)$$

Retarded propagator type (causality as proof argument):

$$G_R : (t \ t' : \text{Time}) \rightarrow (t' < t) \rightarrow \text{Sp} \rightarrow \text{Sp} \rightarrow \text{Field}$$

BFSS cortex (coordinates emerge from eigenvalues):

$$\text{CortexCoords} \equiv \text{eigenvalues}(X), \quad X^\dagger = X, \quad X \in M_{3 \times 3}(\mathbb{C})$$

Full USF type:

$$\text{USF} \equiv \sum_{(\sigma : \text{Scale}_{20})} (\text{Sub}(\sigma) \times G_R(\sigma))$$

III • Attractor Dynamics

Hopfield energy:

$$H(\mathbf{e}) = -\frac{1}{2}\mathbf{e}^\top W \mathbf{e}$$

FM-HN inverse temperature (Lean 4 verified):

$$\beta(t) = \beta_0 + \kappa |\Phi(t)|^2 \xrightarrow{\Phi \rightarrow 0} \beta_0$$

Consciousness threshold T_c : spectral gap opens at $|\phi| > T_c$

Kramers mean first-passage time:

$$\langle T \rangle = \frac{2\pi}{\omega_A \omega_B} e^{\Delta V/D}$$

Trauma asymmetry: $T_{\text{form}} \ll T_{\text{dissolve}}$ because crossing barrier from above vs below

WKB tunnelling amplitude:

$$P \approx \exp\left(-\frac{2}{\hbar_{\text{geo}}} \int_{q_1}^{q_2} \sqrt{V(q) - E} dq\right)$$

Arnold tongue (Huygens locking, width = social/attentional bandwidth):

$$|\omega_{\text{ext}} - \omega_0| < \Delta\omega_{\text{lock}}(\kappa)$$

SHO frequency (physlib ω_{sq}):

$$\omega^2 = k/m \quad (\text{dispersion: } \omega^2 = v^2 k^2)$$

IV • 11D Decomposition

$$M_{11} = M_4 \times X_7, \quad X_7 = D_{5-7} \times D_8 \times D_{9-11}$$

- D_{1-4} **Spacetime** — body in 3+1D Lorentzian space
- D_{5-7} **Propagator** — somatic EM field / D3-brane gauge field
- D_8 **Limbic** — Hořava-Witten orbifold $S^1/\mathbb{Z}_2$, $[-1, +1]$
- D_{9-11} **Cortex** — BFSS matrix eigenvalues (emergent geometry)

Organism hierarchy:

$$11\text{D} \xrightarrow{\text{project}} 7\text{D} \xrightarrow{\text{project}} 4\text{D}$$

V · Key Identifications

- **SHO** $\equiv$ **Green's fn** — string vibration = field impulse response
- **Nash** $\equiv$ **Hopfield min** — $\text{Nash}(G) = \arg \min H(\mathbf{e})$
- **Rights** $\equiv$ **topo. invariants** — stable under cts. landscape deformation
- **Rule of law** $\equiv$ **ergodicity** — $\bar{f}_{\text{time}} = \bar{f}_{\text{ensemble}}$
- **Rapport** $\equiv$ **Huygens locking** — $\dot{\phi}_1 - \dot{\phi}_2 \rightarrow 0$
- **SQ** $\equiv$ **spectral gap** — $\lambda_1(\hat{L}_{ij})$
- **Crash** $\equiv$ **phase transition** — $T \rightarrow T_c^+, \xi \rightarrow \infty$
- $b \approx 1 \equiv$ **universality** — Gutenberg-Richter from symmetry group
- **Groove** $\equiv$ **Arnold tongue entry** — η_{lock}
- **Somatic therapy** — $\eta_{\text{som}}/\eta_{\text{cog}} \approx \kappa_{\text{EC}}^{-1}$

VI · Clinical Operators

ASC — high β , narrow Arnold tongue, deep stable attractors

CPTSD — non-ergodic field, $\kappa_{\text{EC}} \ll 1$ (EC decoupled)

ADHD — high field temperature T , flat landscape, rapid transitions

Co-occurrence ($\text{ASC} \cap \text{CPTSD}$): high $\beta \rightarrow$ deeper trauma wells on adverse input

Window of tolerance = temporal bandwidth of G_R :

$$T_{\text{field}} \in [T_{\text{min}}, T_{\text{max}}] \Leftrightarrow \text{processing possible}$$

Somatic injection bypasses EC junction, acts directly on Φ

VII · Scale Map (20 levels, $\sigma \in \{0, \dots, 19\}$)

- σ 0–2: Quantum/atomic — $\hbar_{\text{geo}}$, nuclear
- σ 3–5: Molecular/cellular — ion channels, neurons
- σ 6–7: Organism — whole brain, somatic field
- σ 8–9: Dyadic/group — rapport, swarm $O(N^2)$
- σ 10: Geographic/geological — dialect spread, tectonics
- σ 11–14: Social/civilisational — culture, economy, law
- σ 19: Cosmological — CMB, $\Lambda \equiv \langle \text{tr } \Phi \rangle_0$

Cosmological limit: $\kappa_{\text{bio}} \rightarrow 0 \Rightarrow$ linearised Einstein eq.

VIII · Lean 4 Status (2026-08-09)

- MTheoryIsomorphism $\omega^2 = k/m$ ✓ physlib trajectory_equationOfMotion
- MTheoryIsomorphism wave equation ✓ physlib planeWave_waveEquation
- LimbicHopfield FM-HN Correspondence ✓ kernel
- SwarmPropagator $O(N^2)$ bound ✓ kernel
- LimbicTunnel WKB amplitude $\Theta(W)$ ✓ native_decide
- ScaleUniverse consciousness threshold ✓ kernel
- BFSSIsomorphism cortex eigenvalue emergence ✓ Mathlib
- G holonomy of X_7 □ axiom (needs Riemannian geometry)
- Zoom Operator covariance □ axiom (needs tensor field formalism)

GitHub: ITI-Theory/U · Zenodo: doi.org/10.5281/zenodo.20460771